

Experimental investigations of the acoustic backscattering of a single cavitation bubble

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Abstract

There is a great interest in the oscillation behaviour of a single cavitation bubble in a sound field. Bubble dynamics is investigated by evaluating the acoustic backscattering of a high-frequency image sound field from the bubble. It is shown that the scattering technique permits the simultaneous evaluation of the amplitudes and of a shift in frequency in the scattered image sound field to obtain information about bubble dynamics. Thus, an acoustic observation of the bubble oscillation is performed. © 1997 Elsevier Science B.V.

Keywords: Bubble oscillation; Scattering; Single bubble cavitation

1. Introduction

The oscillation behaviour of cavitation bubbles has been the subject of investigation for a long time. In recent years, a lot of effort has been put into the evaluation of single-bubble cavitation phenomena. Gaitan et al. [1] performed experiments to generate single-bubble cavitation; in this way, they were able to detect the phenomenon of single-bubble sonoluminescence. In their work, and in the subsequent work of other groups performing single-bubble cavitation experiments to determine bubble dynamics, optical methods have been predominantly used for observation of oscillating bubbles. In addition to directly observing the bubble with very fast cameras (CCD- or streaker-camera), the principle of laser scattering has been applied to obtain information on bubble dynamics.

Evaluation of the acoustic scattering behaviour of the cavitation bubble [2,3], in addition to optical methods, is suitable for the investigation of bubble oscillation. A high-frequency image sound field can be scattered by the bubble, while evaluating the scattered image sound contributions permits one to obtain information about the oscillation behaviour of the bubble. This paper reports on the investigation of the backscattering behaviour of a single cavitation bubble from the viewpoint of

observations of bubble oscillations. For this reason, a bubble oscillating in a primary driving sound field is exposed to a second high-frequency image sound field, as shown schematically in Fig. 1. The scattered sound contributions are received and evaluated in two different ways.

After a short summary of the theoretical aspects of bubble oscillations and sound radiation of a cavitation bubble, the influence of the image sound on the bubble dynamics is investigated theoretically. The experimental investigations start with the description of the experimental set-up used to perform single-bubble cavitation. Two ways are possible in evaluating the backscattered high-frequency sound. The amplitudes of the backscattered sound provides information about the momentum radius of the bubble. On the other hand, due to the oscillation of the bubble as secondary source, a shift in frequency in the scattered sound appears giving the bubble wall velocity. Experimental results are presented and compared with a numerical calculation of the bubble radius versus time.

2. Theoretical considerations

To obtain an estimation for the backscattering behaviour of a cavitation bubble and to calculate the influence of the additional high-frequency image sound field on

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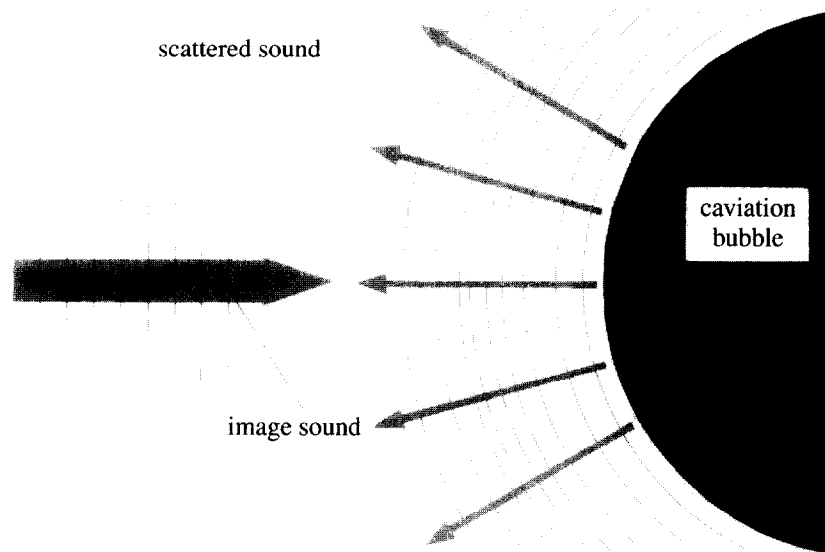


Fig. 1. Backscattering of a high-frequency image sound by the bubble.

the original bubble oscillation, a single bubble in an indefinite fluid is assumed to radially oscillate in a sound field. This sound field for driving and generating the single cavitation bubble is called the primary sound field.

Under these assumptions, bubble behaviour can approximately be described using the Gilmore equation [4]. Among other effects, the model considers radiation losses and the change in the propagation velocity of sound next to the bubble due to the flow around the oscillating sphere. As an accurate description of the thermal gas behaviour inside the bubble, Prosperetti worked out a combined numerical solution of the equation of motion together with the equations of enthalpy, continuity and momentum [5]. However, in the case considered experimentally for demonstration purposes with relatively large bubbles in stable oscillation, the polytropic approach serves as a good approximation to obtain an estimation of the bubble dynamics.

If, for example, a bubble with an equilibrium radius of $30\text{ }\mu\text{m}$ is exposed to a primary excitation sound field with a frequency $f_p = 20\text{ kHz}$ and a driving pressure $p_p = 140\text{ kPa}$, the characteristic oscillation behaviour is shown in the radius function of time $R(t)$ in Fig. 2. Of course, in the range of such high-pressure excitation, violent bubble collapses arise. To observe this bubble acoustically, a second high-frequency image sound field is applied, with an amplitude much weaker than the primary one. The image frequency range has been chosen between 22 and 30 MHz to minimize the influence of the image sound on the oscillation combined with a good resolution and a low cavitation noise level in this range, which disturbs the scattering measurements. In numerical calculations, the radius function of time has been calculated when excited by the primary and the image sound field. To perform realistic numerical

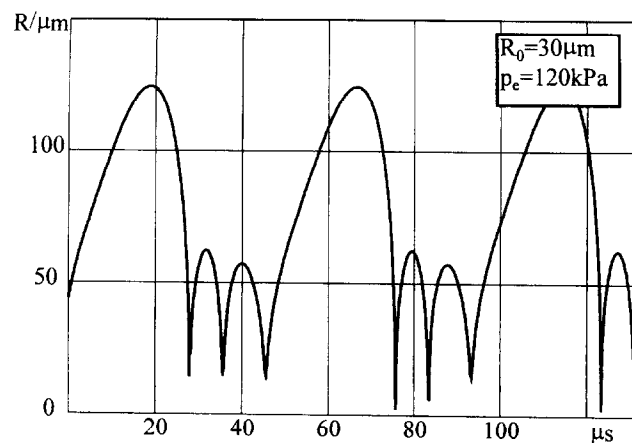


Fig. 2. Radius of an oscillating bubble calculated numerically.

calculations as in the experimental case, image excitation bursts of a certain length with a high-frequency carrier have been used.

First, the radiation of sound by the bubble is considered. Based on the Gilmore approach for bubble oscillation, the sound radiation of the bubble can be determined by the approach of Gilmore and Akulichev [6]. The radiated sound contains contributions in the whole frequency range. The sound radiation due to the primary driving field is predominantly in the lower frequency range up to 1 MHz, but even at higher frequencies, sound is radiated. During the collapse phase, for a violent collapse in particular, shock waves are emitted with a spectral content spreading equally over a wide frequency range. The additional image sound field is responsible for radiated components mainly in the range of the image frequency. To evaluate the scattered image sound, band filtering helps to cut

out the sound radiated by the bubble due to the image excitation and diminishes the effects of the other radiated components. The sound radiation has been calculated for two different amplitudes of the primary pressure with a pressure of the image sound $p_i = 70$ kPa. In Fig. 3, the radiated sound band filtered around the image frequency is demonstrated. In the first case [Fig. 3(a)], a primary excitation amplitude of $p_p = 70$ kPa has been chosen. In this case, a stable bubble oscillation develops. In Fig. 3(b), the primary pressure amplitude was increased to $p_p = 140$ kPa. Here, violent collapses can be observed. As a comparison, in Fig. 3 the radius functions of time as calculated by the Gilmore equation are given. From the view of the image frequency, which is three orders of magnitude higher than the primary driving field, the bubble pulsation is quasi-static during one image cycle. For a bubble driven this way, the

returned signal is a measure for the scattering – and thus for the geometric – cross-section of the bubble, which varies with the bubble oscillation, as has been stated by Leighton and others [7,9]. The image frequency is orders of magnitude higher than the resonant frequency of the bubble and the primary driving field. Obviously $R(t)$ corresponds to the envelope of the scattered sound band filtered around the image frequency. Various numerical calculations have been performed for different equilibrium bubble radii leading to similar results. Limits of these predictions may appear if the wavelength of the image sound is of the same range as the bubble size. Non-radial oscillations develop: their effects on the scattering behaviour are discussed later.

Deviations from the proportionality of the bubble radius to the scattered image pressure appear in the second case of a high primary pressure. During the phase of the violent bubble collapse, peaks in the band filtered scattering contributions appear. These peaks are linked to the emission of shock waves during the collapse phase. Thus, during a violent collapse, the backscattered contributions do not give information about the momentum bubble radius. In the case of a low primary excitation, there is no distortion over the whole oscillation period due to shock wave emission.

If the principle of acoustic backscattering is used to observe the bubble oscillation, there should be no or only negligible distortions of the original bubble motion due to the measurements. The additional image sound field is expected to influence the dynamic processes because the bubble radiates the high-frequency image sound as a secondary source. To get an estimation of the change in the bubble motion, first the radius functions of time have been calculated for primary excitation only. In comparison, the additional image sound burst is applied. As an illustration of the driving field, the primary excitation (a) is demonstrated in Fig. 4. A burst of a high-frequency image sound (b) is overlaid; its amplitude has been chosen as $p_i = 70$ kPa, i.e. a very high amplitude, to point out the effects of the image sound field. Of course, an image sound of this pressure is unrealistic and applied for demonstration purpose only. The burst length was a little longer than one period of the bubble oscillation period. Again, in the first case, the primary amplitude was defined to be $p_p = 70$ kPa and both excitation pressures are of the same amplitude. In Fig. 4(c), the difference between the undisturbed and the disturbed bubble radius is demonstrated for an equilibrium radius of $30\text{ }\mu\text{m}$. The absolute value of the deviations maintains beyond 45 nm so the relative difference is in the range of one thousandth of the equilibrium radius. After some periods of bubble oscillation, the effects of the image sound dissolve and the bubble moves as in the undisturbed case. If the primary amplitude is increased to a value of

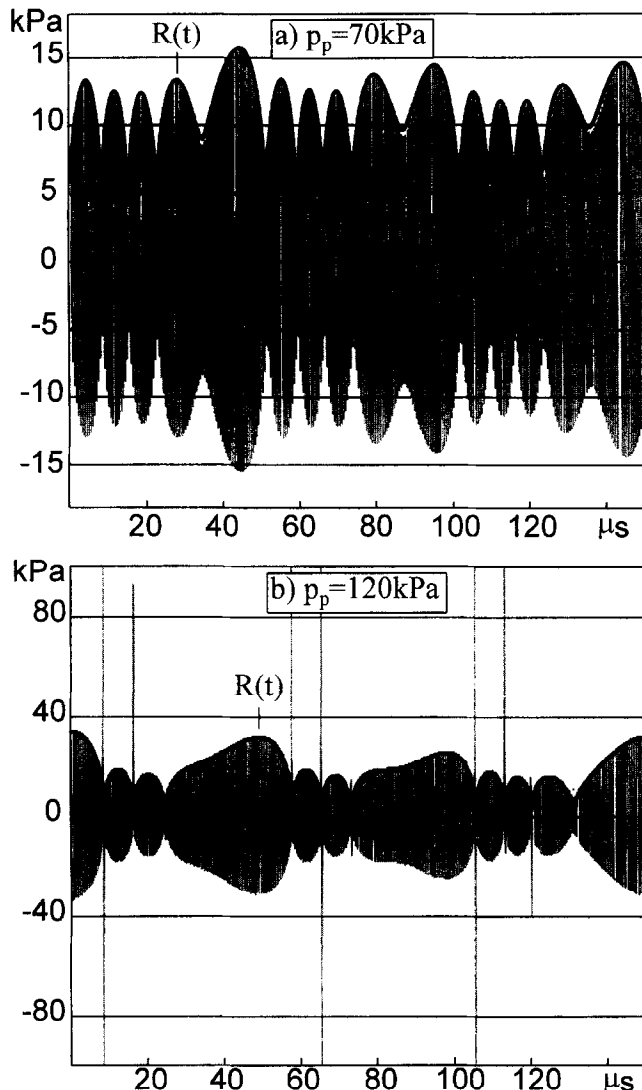


Fig. 3. Sound radiation of a bubble driven by two frequencies, $R_0 = 30\text{ }\mu\text{m}$, $f_p = 20\text{ kHz}$, $f_i = 22\text{ MHz}$, $p_i = 70\text{ kPa}$: (a) $p_p = 70\text{ kPa}$; (b) $p_p = 120\text{ kPa}$, compared with the theoretical bubble radius $R(t)$.

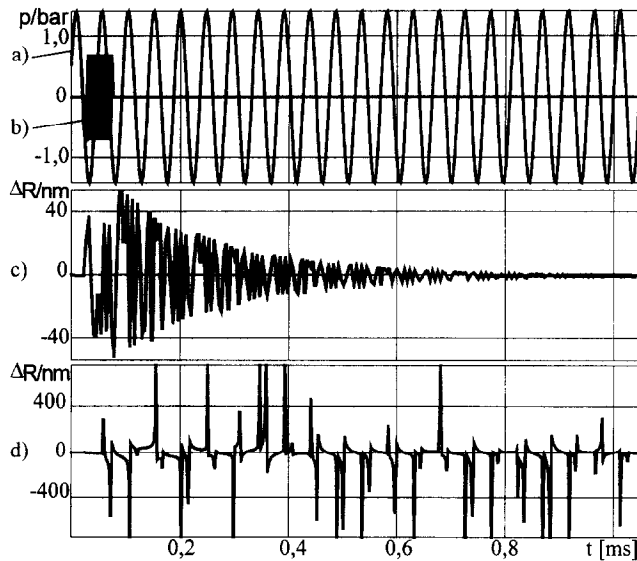


Fig. 4. (a) Primary $f_p=21$ kHz; (b) burst of the image sound $f_i=22$ MHz, $\bar{p}_i=0.7$ bar; deviations of the bubble radius due to image sound with (c) $\bar{p}_p=0.7$ bar; and (d) $\bar{p}_p=1.4$ bar.

$p_p=140$ kPa, the influence of the second field on the bubble increases. As demonstrated in Fig. 4(d), deviations of up to 4% appear for this case of bubble oscillation. Numerical calculations for some periods of the primary excitation give a longer decay time for the deviations in the bubble radius. Although this decay may be overestimated due to the small degree of damping incorporated in the polytropic approach for the thermal behaviour inside the bubble, the characteristic development is pointed out. Obviously the differences, ΔR , become worth mentioning only for very short periods of time and an exact evaluation leads to the phase of violent bubble collapse. During the rest of the bubble oscillation, the deviations are again very small. As mentioned above, the scattered signals cannot be used during the phase of the violent collapse as due to their small value compared with other experimental faults, the remaining deviations are neglected in the experimental investigations; in particular, when only small image amplitudes are applied.

In comparison with the theoretical studies described below, numerical calculations have been performed on the Gilmore approach, while the assumptions of Prosperetti's nearly isothermal behaviour inside the bubble were incorporated [5]. Further investigations of the bubble oscillation using a slightly modified form of the Keller–Miksis equation [8] were carried out, which in opposition to the Gilmore equation, assumes a constant sound speed in the fluid. Apart from more damping in the nearly isothermal approach, the results of both investigations look very similar to those already presented. Thus, the numerical calculations based on the Gilmore model with the polytropic approach for the gas inside the bubble, including the Gilmore–Akulichev

assumptions of sound radiation, seem to be a sufficiently exact estimation for the high-frequency backscattering behaviour of the cavitation bubble in the considered range of bubble oscillation.

While for the primary excitation, the bubble is much smaller than the wavelength and the radially oscillating bubble is an appropriate assumption, this condition is no longer fulfilled for the wavelength of the image sound field. During the excitation of the bubble with the high-frequency image sound field, the wavelength is of the order of the bubble size, so non-radial oscillations may appear in addition to the radial modes of oscillation. The sound emitted by the bubble will be composed of the contributions emitted in radial and non-radial oscillations. These sound components are differently dependent on the distance from the receiver to the bubble [9]. Looking from the far field, the dominating components will be radiated by the radial modes of oscillation, so the assumption of the radially pulsating bubble is justified in the theoretical considerations. The exact evaluation of this aspect will be subject of future work.

3. Experimental investigations

The bubble oscillating in the primary sound field being exposed to a burst of a high carrier frequency (22–30 MHz) is pointed out in Fig. 5. The scattering behaviour has been investigated in the backscattering direction. In the experimental procedure, it is useful to apply a single transducer to transmit and to receive the high-frequency field. This way, the adjustment of the image transducer on the cavitation bubble is simplified. The transition time of the image sound from the transducer to the bubble and back defines a limiting condition for the image burst length and the transducer is not able to send and to receive a signal at the same time. In the experiments, the geometrical arrangement allowed to use image bursts with a duration of 52 μ s to observe a little more than one period of the bubble oscillation in one measuring cycle, as mentioned in the advanced theoretical studies.

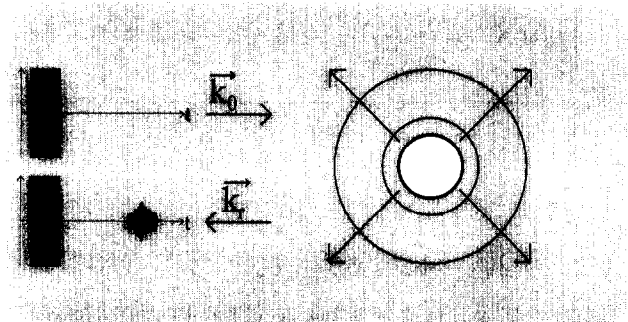


Fig. 5. Schematic situation of backscattering.

The experimental investigations have been performed in a common arrangement for generating single-bubble cavitation, already described in Refs. [1,9]. The schematic set-up displayed in Fig. 6 consists of a cylindrical body built from a piezoceramic ring fixed to a ring of glass. The top and the boundary in the bottom behave like soft surfaces to the sound such the cavitating bubble is positioned in the antinodes of the pressure field. In the bottom of the cylinder, a high-frequency PVDF transducer has been mounted in such a way that its focus coincides with the bubble position. The radiation of the image signal is controlled by a programmable delay circuit triggered on the primary excitation. An image burst is radiated into the tube and the backscattered signals are received using a separation network. The measuring data is digitized and transferred to a personal computer for further processing. Due to the relatively wide focus of the image transducer a little shift in the bubble position may not dramatically disturb the experimental results.

A typical signal structure as determined from the experiments can be seen in Fig. 7. Here the received sound is displayed over three periods of bubble oscillation. Among the backscattered image sound, a typical cavitation noise is also contained. The primary driving field was shielded by high-pass filtering. In order to separate the backscattered image sound, the measuring signal was band filtered around the image frequency

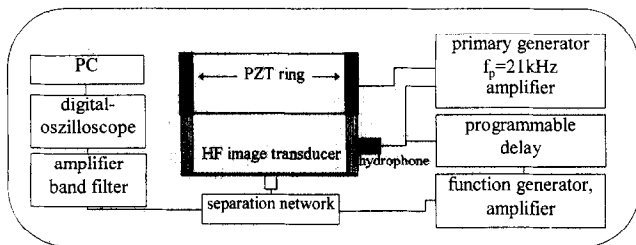


Fig. 6. Experimental set-up for generating single-bubble cavitation.

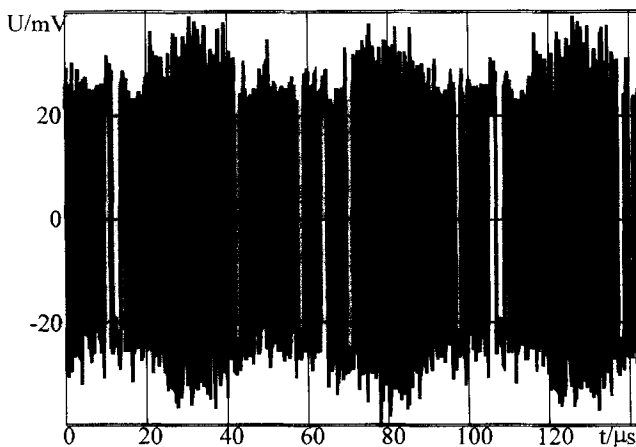


Fig. 7. Signal received by the image transducer for the periods of bubble oscillation.

according to the theoretical studies. Following this, the envelope of the backscattered sound was calculated exploiting the Hilbert transform to obtain a function proportional to the radius function of time. Fig. 8 shows the experimental data from the previous Fig. 7, extracted in this way. The result is a curve corresponding to the time-dependent function of the bubble radius. The experimental data is obtained on a cavitation bubble oscillating in a primary sound field of $p_p = 65$ kPa. The data is not averaged and the results can be reproduced. A comparison of the results obtained by evaluation of the scattering amplitude with a numerical calculation of a bubble oscillation for the same driving sound field and approximately the same average size is demonstrated in Fig. 9. The evaluation of the backscattered image sound amplitudes is a reliable and exact method to observe the dynamic behaviour of the bubble oscillation.

To prove the experimental results, additional optical investigations with a camera have been performed. This

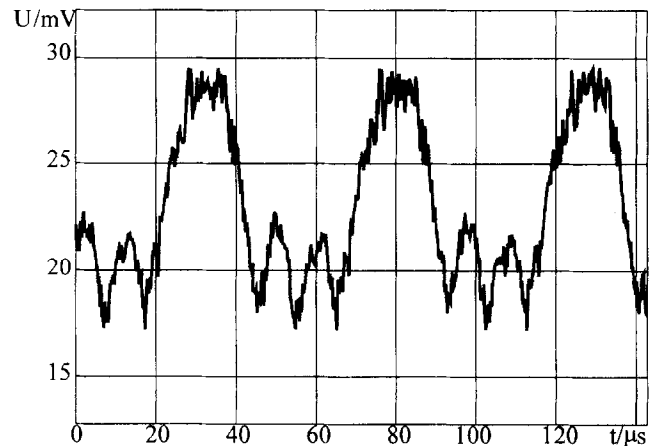


Fig. 8. Envelope of the band filtered backscattered signal.

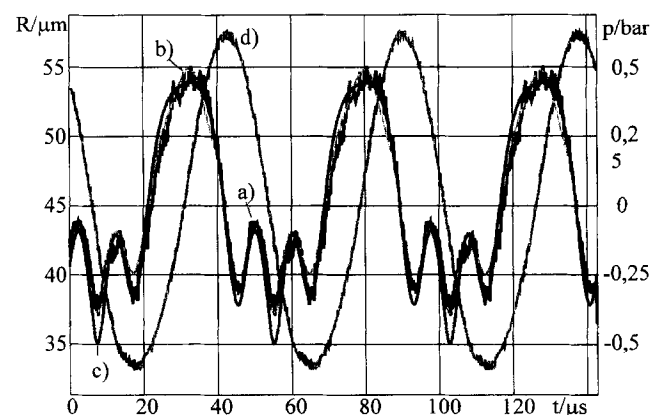


Fig. 9. Radius functions of time: (a) amplitude evaluation; (b) evaluation of the frequency shift; (c) Gilmore $R_0 = 42$ μm , $p_p = 0.6$ bar; and (d) experimental sound pressure.

way the acoustic backscattering technique can be calibrated and an average bubble radius is found. By this method, the value obtained is used to process the bubble radius function of time in the evaluations of the spectral characteristics of the backscattered image sound, which is described below.

Consider the cavitation bubble oscillating in the primary sound field. When exposed to the image sound, the bubble radiates the scattered high-frequency signal. The bubble surface is a secondary source for the scattered sound. Because of the oscillating bubble, from the observers point of view, the bubble wall represents a moving source. As is well known, a moving source accounts for a shift in frequency between the radiated sound and the received signals. Assuming a bubble wall velocity smaller than the speed of sound in water, the frequency shift can be evaluated using a linearized Doppler approach. This way the relationship between the shift in the image frequency and the bubble wall velocity is given as connected by the speed of sound in water – assumed to be constant in the considered range of the bubble oscillation – and the frequency of the image sound field. If the experimental data from Fig. 7 is band filtered around the image frequency to reduce the disturbances coming from the primary excitation and the broadband cavitation noise in the received scattering signals, the frequency deviations from the original image frequency can be determined. This was carried out by calculating the momentum frequency of the backscattered image sound. For this reason, the time derivative of the phase of the backscattered contributions was taken to get the momentary frequency [10]. The phase has been determined by the application of the measuring signal and its Hilbert transform, already calculated to obtain the envelope of the backscattered signals for the evaluation of the scattering amplitudes. After narrow band filtering, the momentary frequency reflects the frequency of the backscattered image sound. In Fig. 10, the shift in frequency extracted by means of

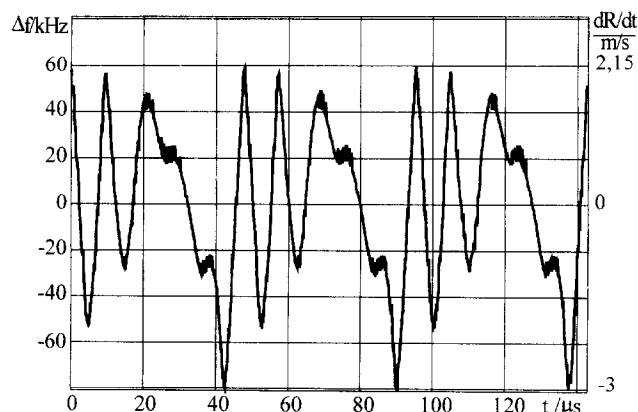


Fig. 10. Deviations of the scattered frequency from the image frequency.

the momentary frequency of the experimental data compared with the image frequency is shown. Due to the coupling between the frequency shift and the motion of the bubble surface, the bubble wall velocity is only approximately given, as demonstrated in the same figure. In the case investigated in our experimental data, the deviations between the image frequency and the backscattered frequency take values up to 80 kHz, which can be deduced to a bubble wall velocity of up to $\dot{R} = 3$ m/s. This result again predicts a stable bubble oscillation with the velocity of the bubble surface low compared with the speed of sound.

Time integration of the bubble wall velocity $\dot{R}$ provides the curve of the bubble radius during the oscillation. Based on a known average radius of the cavity as integration constant, which may be obtained in optically, the radius as a function of time can be calculated. The average bubble radius was determined by the use of a fast camera or by microscopic investigations of the oscillating bubble. In the example in Fig. 7, the average radius was determined to be 40–45 μm .

Fig. 9 gives an overview over the experimental results obtained by acoustic scattering. Trace (d) is the sound pressure of the primary sound field as monitored by a hydrophone in order to get a reference for the phase and amplitude of the driving sound field, while high-frequency components are shielded by a low-pass filter. Line (a) again presents the curve from Fig. 8, whilst in curve (b), the results achieved from the evaluation of the spectral shift in the scattered image sound are shown. Both experimental results are plotted without averaging. In comparison, in trace (c), there is the radius function of time calculated numerically under the same conditions as in the experimental investigations. Obviously there are good agreements between the radius curves. The two ways to process the experimental data permit one to obtain independent information of the bubble oscillation at the same time. This redundant knowledge can be used to minimize measurement uncertainties. If the bubble position is not ideally fixed, i.e. in a set-up as in Fig. 6, the lateral bubble motion gives an additional frequency shift. This distortion is detected in Fig. 9 in the mismatch of both experimental curves. Applying the backscattering principle perpendicular to the middle axis of the experimental arrangement permits the protection of the experimental data from this influence.

In the range of a more intense bubble oscillation, the bubble wall velocity may reach or exceed the sound speed during the phase of a violent collapse. If the condition that the bubble wall velocity is smaller than the speed of sound is not fulfilled, the evaluation using the Doppler approach cannot be applied any more. In any case, due to the high-frequency contributions of the shock waves emitted during a violent bubble collapse, the backscattered sound cannot even be resolved during these phases of bubble oscillation.

4. Conclusion

In addition to optical observations of the dynamic behaviour of a cavitation bubble, the acoustic scattering technique is an exact and reliable way to observe oscillating bubbles. Also, with the examination of the back-scattering amplitudes of a high-frequency image sound, a shift in frequency in these sound contributions can be detected, which leads to redundant information about the momentum radius function of time. However, the spectral evaluation of the scattered signals provides information about bubble oscillation. Apart from the application of this method in investigations of the acoustic scattering, spectral evaluations will even be fruitful in optical scattering experiments. This way, problems occurring due to changing refraction indices inside the bubble during the collapse phase and estimations of the temperature raise inside the bubble could be investigated.

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